

EEG-EDA Analysis Report

VR Stimulus Analysis for EEG Data

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This script analyzes EEG recordings from VR headset sessions with different stimulus periods.

Stimulus Pattern (Total duration: 3 minutes 20 seconds / 200 seconds):

- 0-20 sec: white neutral
- 20-50 sec: warm color
- 50-80 sec: cold color
- 80-110 sec: warm color
- 110-140 sec: cold color
- 140-170 sec: warm color
- 170-200 sec: cold color

The goal is to compare EEG data between warm and cold color stimulus periods and create visualizations to show differences in brain activity.

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Analysis Parameters

Parameter	EEG	EDA
Sampling Rate	250 Hz	500 Hz
Window Length	1 second (250 samples)	1 second (500 samples)
Window Overlap	96% (240/250 samples)	90% (450/500 samples)
	Notch Filter (50 Hz, Butterworth)	Optional smoothing; resampled and baseline corrected
Filters Applied	Bandpass Filter (0.5-40 Hz, 4th order Butterworth)	
Frequency Bands	Delta (0.5-4 Hz), Theta (4-8 Hz), Alpha (8-13 Hz), Beta (13-30 Hz), Gamma (30-40 Hz)	Not applicable
	Time-Frequency CSD, Max (30-40 Hz)	Tonic (SCL), Phasic (SCR peaks), peak count, amplitude, IQR, SD
Features Extracted	Freq: Band powers, ratios, spectral entropy, flatness	
Spectral Methods	Welch's PSD (nperseg=250), Periodogram with high overlap, Spectral edge frequency, coherence	Welch's PSD (nperseg=500), enhanced SCR periodogram, SCL smoothing (4s window)
Cross-Channel Analysis	Mean coherence between all electrode pairs	Not applicable
Classification	SVM (RBF kernel) with z-score normalization	Integrated features with EEG, trained via user-labeled states (Relaxed, Focused, Stressed)
Topographical Mapping	10-20 standard montage (Fp1-P4) with 2D interpolation and dynamic visualization	Not applicable

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Analysis Results

Classification Performance

Model Type: Support Vector Machine (RBF kernel)

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Spectral Analysis Results

EEG Power Spectral Density

EEG spectral results not available

EDA Component Analysis

EDA spectral results not available